Supplementary material: Search strategy

Graph Neural Networks for $PM_{2.5}$ Forecasting: Review on Recent Advancements, Challenges, and Outlooks

Table S1: Complete database-specific search strategies documented for the systematic review (2019–2026). Entries reproduce the authors' recorded Boolean concept groups and source-level results; the four searches generated 2,180 exported records before cross-database deduplication.

Database / run	Recorded search expression	Before filters	After filters	Removed
Scopus (S1.1)	TITLE-ABS-KEY((graph neural network* OR graph convolutional network* OR graph attention network* OR spatiotemporal graph neural network* OR spatio-temporal graph neural network* OR spatial-temporal graph neural network* OR dynamic graph neural network* OR adaptive graph neural network* OR graph transformer* OR multi-graph neural network*) AND (PM2.5 OR PM 2.5 OR fine particulate matter OR air quality OR air pollution OR air quality index OR AQI) AND (forecast* OR predict* OR estimat* OR analy* OR assessment))	449	391	58

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Supplementary Table S1 (continued)

Database / run	Recorded search expression	Before filters	After filters	Removed
ScienceDirect (S1.2)	(graph neural network OR graph convolutional network OR graph attention network OR spatiotemporal graph neural network) AND (PM2.5 OR air quality OR air pollution) AND (forecasting OR prediction)	1,599	1,092	507
ScienceDirect (S1.3)	(dynamic graph neural network OR adaptive graph neural network OR graph transformer OR multi-graph neural network) AND (PM2.5 OR air quality OR air pollution) AND (forecasting OR prediction)	108	81	27

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Supplementary Table S1 (continued)

Database / run	Recorded search expression	Before filters	After filters	Removed
SpringerLink (S1.4)	(graph neural network OR graph convolutional network OR graph attention network OR spatiotemporal graph neural network OR spatio-temporal graph neural network OR dynamic graph neural network OR adaptive graph neural network OR graph transformer OR multi-graph neural network) AND (PM2.5 OR PM 2.5 OR fine particulate matter OR air quality OR air pollution OR air quality index OR AQI) AND (forecasting OR prediction OR estimation OR analysis)	665	616	49
Total	**Four search runs across three databases**	**2,821**	**2,180**	**641**

Search documentation notes.

1. **Source and transcription.** Expressions and counts are transcribed from the supplied *search string – Sheet1.pdf* (one-page search sheet). The expressions are displayed as recorded, without introducing quotation marks, field qualifiers, or new keywords; their exact execution syntax has not been independently verified by rerunning the searches.
2. **Date and filtering.** The sheet labels its coverage "2019–26" and distinguishes results "before filter" and "after filter", but does not identify the precise filter types (e.g., document type, language, access status), individual run dates, or interface versions. The accompanying manuscript reports 5 October 2026 as the overall last search date. These should not be presented as verified run-level settings.
3. **ScienceDirect.** The two recorded search runs yielded 1,092 and 81 post-filter records (total 1,173); possible within-database overlap between search runs cannot be established from this summary sheet alone.
4. **PRISMA accounting.** Scopus contributed 391, ScienceDirect 1,173, and SpringerLink 616 exported records (2,180 in total), confirmed against the original merged-record CSV. The separate cross-database deduplication step removed 243 duplicates, leaving 1,937 records for title screening. The source-specific before/after-filter differences above are not duplicate-removal counts.